Cessna 172 Airfoil

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Introduction

This report presents a computational fluid dynamics (CFD) analysis of the airfoil sections used on a Cessna 172 aircraft. The aircraft's main wing uses a NACA 2412 cambered airfoil, while the horizontal stabilizer uses a symmetric NACA 0012 airfoil. Both profiles were modeled together and simulated in Ansys to study the pressure and velocity fields generated around each surface and to quantify the resulting lift and drag coefficients.

The surrounding air was modeled as an ideal gas with a freestream airspeed of 63 m/s, corresponding to a representative cruise condition for the aircraft. Part A of the assignment establishes the baseline flow field at a 0° angle of attack, while Part B extends the analysis across a range of angles of attack (-2°, 0°, 2°, and 4°) to examine how the aerodynamic coefficients change with pitch.

Geometry, Mesh, and Baseline Flow

The main wing (NACA 2412) and stabilizer (NACA 0012) profiles were generated and placed in a shared computational domain. Each airfoil was surrounded by a structured boundary-layer mesh to resolve the near-wall gradients accurately, transitioning to a coarser unstructured mesh farther from the surfaces to keep the overall cell count manageable. This mesh density was chosen to ensure reliable pressure and velocity data could be collected directly at the airfoil surfaces as well as in the surrounding flow field.

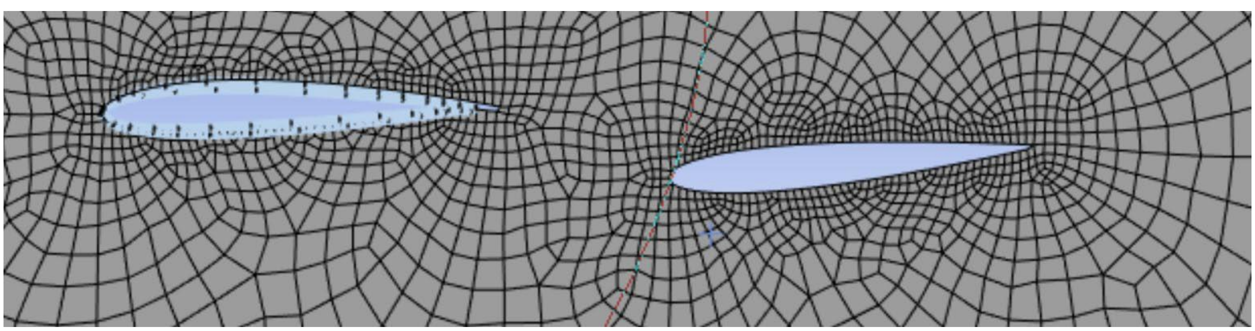

Figure 1. Mesh around stabilizer and main wing

Pressure Field

With the simulation converged, the static (absolute) pressure field around both airfoils was extracted. The results show a region of reduced pressure on the upper (suction) surface of the main wing and a corresponding region of higher pressure on the lower surface. This pressure differential across the wing is the primary mechanism generating lift.

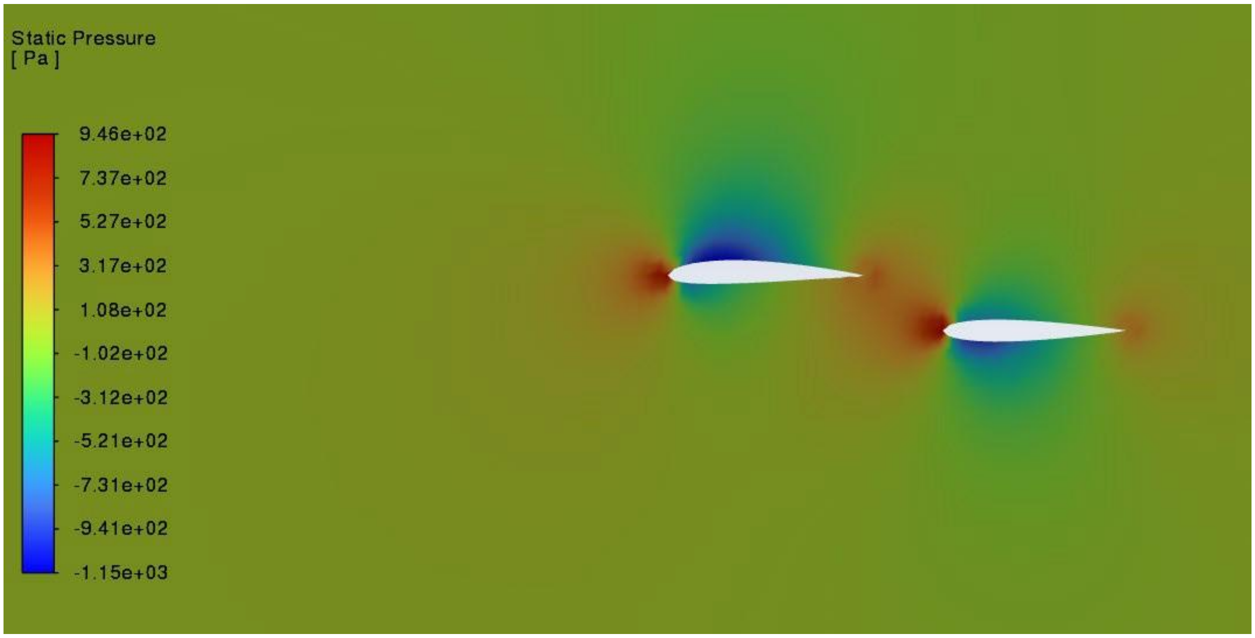

Figure 2. Static Pressure around the wing and stabilizers (Min-1150.4 Pa, Max = 946.3 Pa)

Velocity path lines

Velocity streamlines were plotted around both airfoils to visualize how the freestream flow is deflected and accelerated as it passes each surface. The streamlines bunch closer together above the main wing, indicating locally accelerated flow, which is consistent with the region of lower pressure identified above.

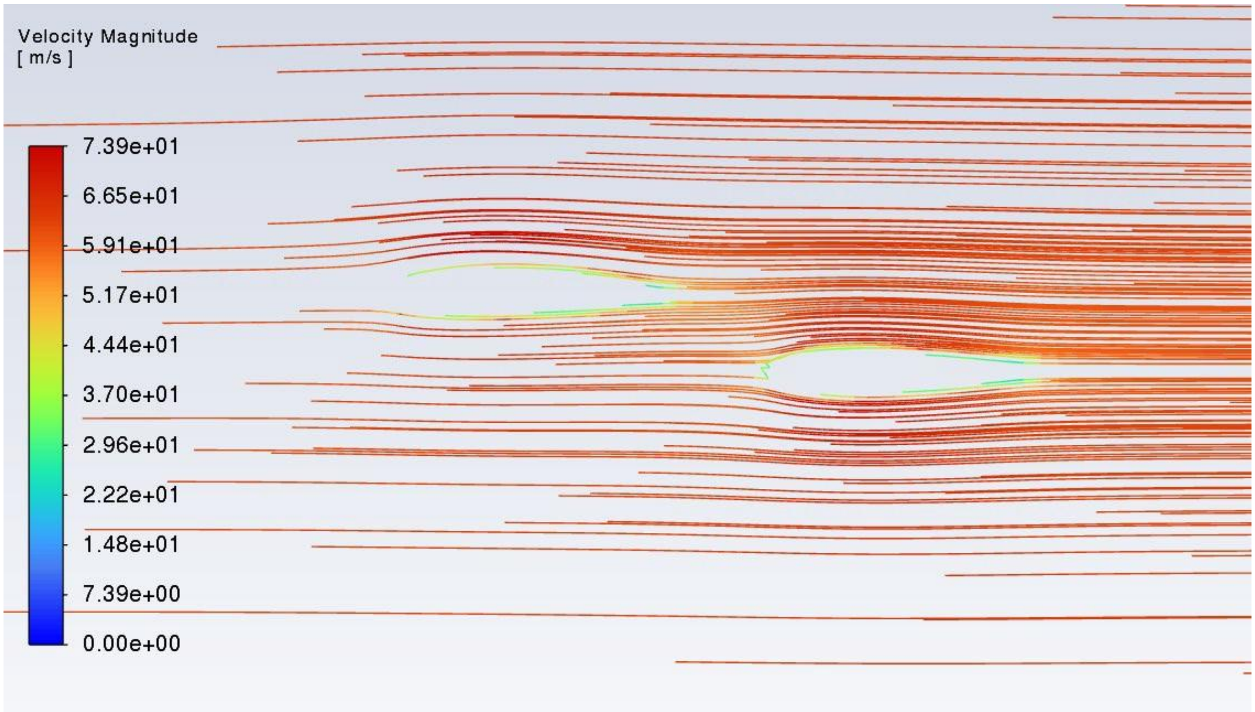

Figure 3. Velocity streamlines round main wing and stabilizer

The velocity magnitude contour reinforces this observation: the flow accelerates most strongly along the upper surface of the main wing, reaching a maximum velocity of approximately 73.9 m/s versus the 63 m/s freestream value. By Bernoulli's principle, this local increase in speed corresponds to the drop in static pressure shown in Figure 2, which is the effect that helps generate lift on the wing.

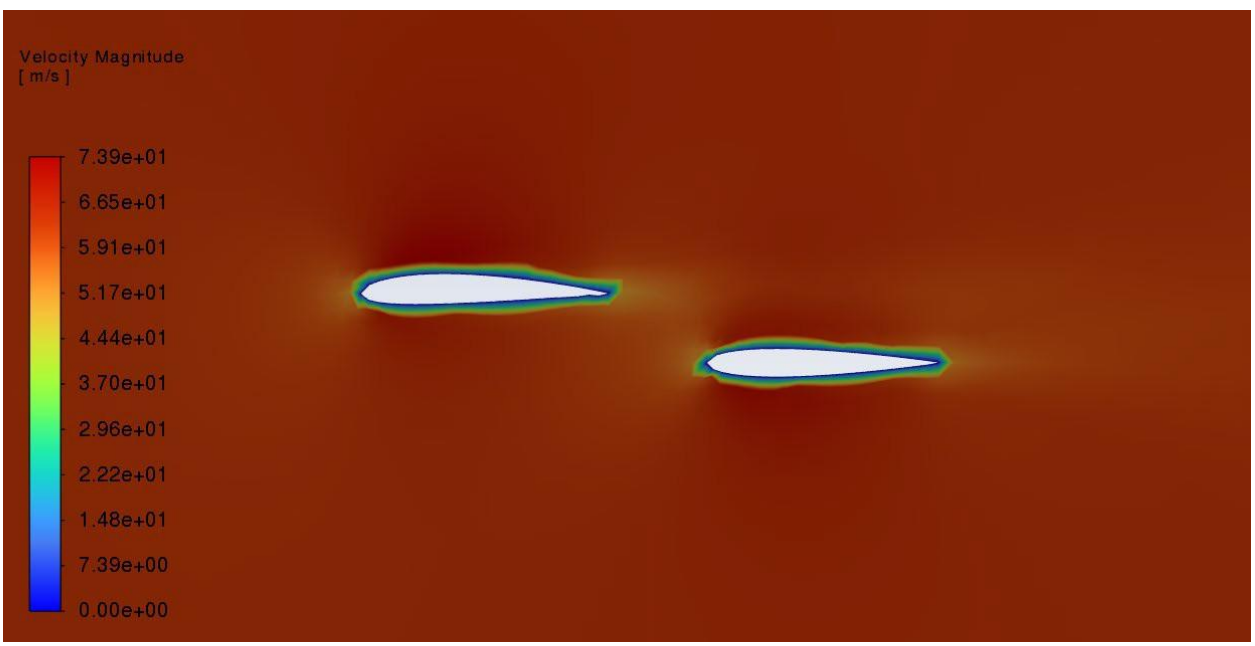

Figure 4. Velocity magnitude contour (0-73.92m/s) around the main wing and stabilizer.

Lift & Drag Coefficient

The lift coefficient (Cl) and drag coefficient (Cd) were monitored over the course of the solution to confirm convergence. At the standard 0° angle of attack on the stabilizer, the lift coefficient converged to approximately Cl = 0.0113, and the drag coefficient converged to approximately Cd = 0.00295. These baseline values serve as the reference point for the angle-of-attack sweep performed in Part B.

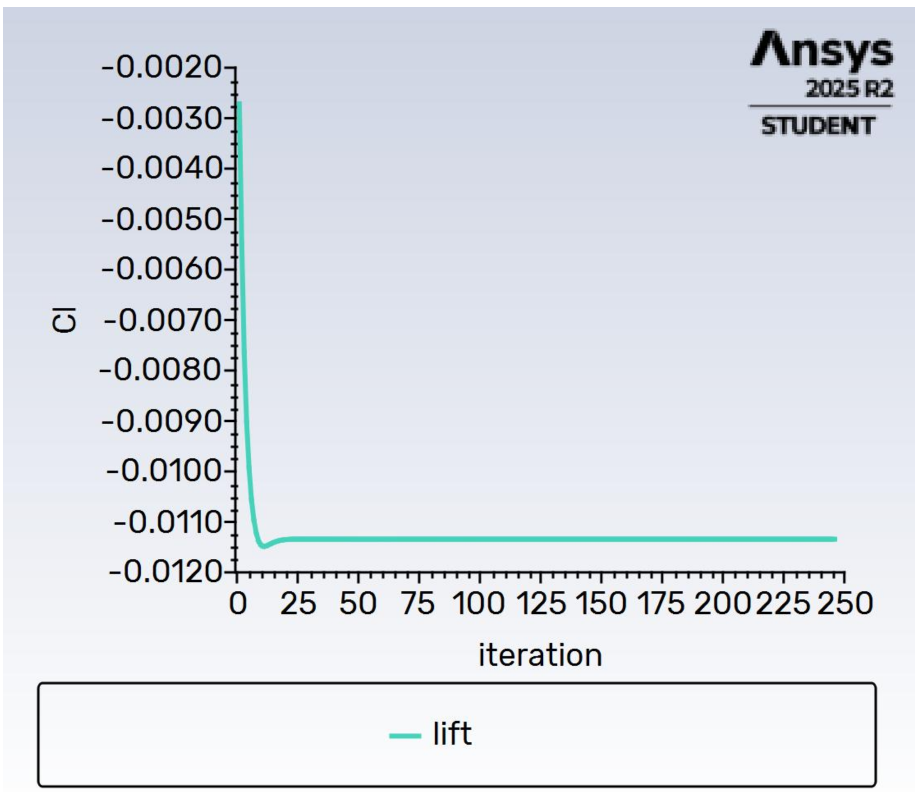

Figure 5. Lift coefficient convergence history, 0 degree angle of attack

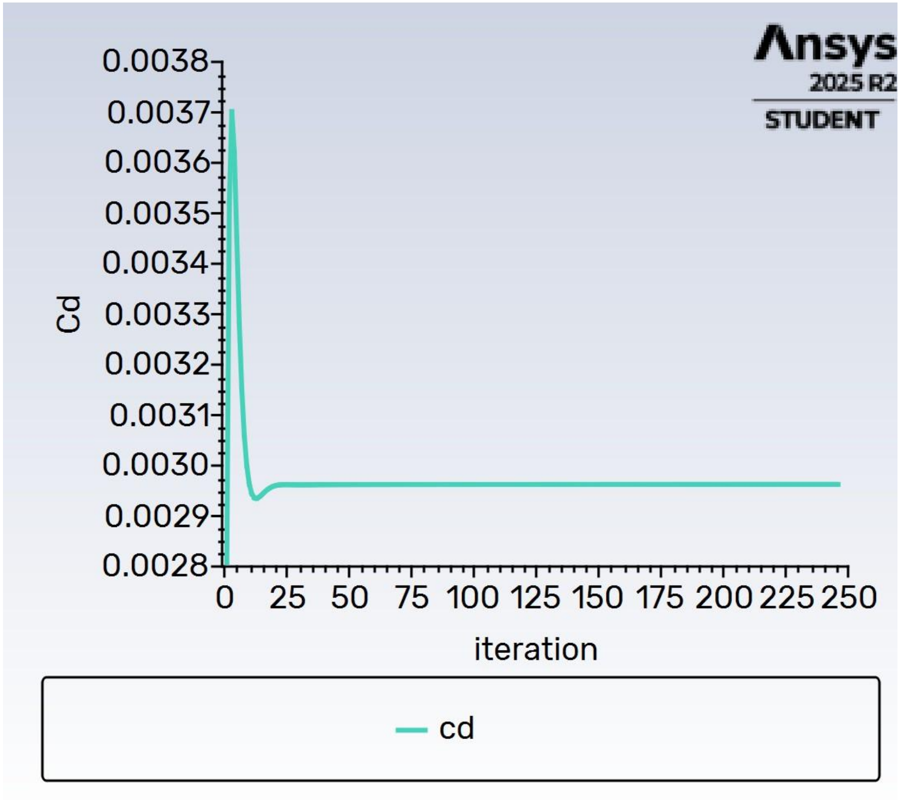

Figure 6 Drag Coefficient convergence history 0 degree angle of attack

Part B: Angle of Attack Sweep

To understand how the aerodynamic performance of the wing changes with pitch, the simulation was repeated at four angles of attack: -2°, 0°, 2°, and 4°. For each case, the lift and drag coefficient histories were monitored until they reached a converged, steady value. The results are summarized in Table 1 and shown individually in Figures 7–10.

Angle of Attack (°)	Lift Coefficient, Cl	Drag Coefficient, Cd	Cl / Cd
-2	≈ -0.0165	≈ 0.00375	≈ -4.4
0	≈ 0.0113	≈ 0.00295	≈ 3.8
2	≈ 0.045	≈ 0.0044	≈ 10.2
4	≈ 0.072	≈ 0.0065	≈ 11.1

Table 1. Converged lift and drag coefficients across the angle-of-attack sweep.

3.2 -2° Angle of Attack

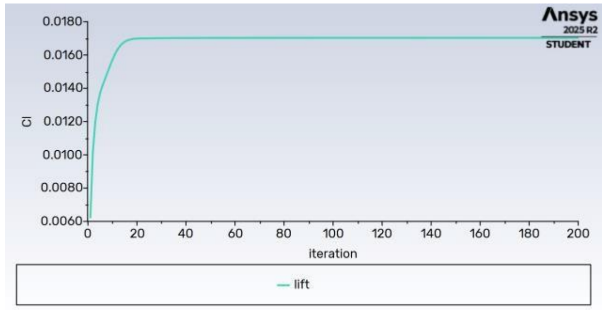

Figure 7a. Lift coefficient, -2° AOA.

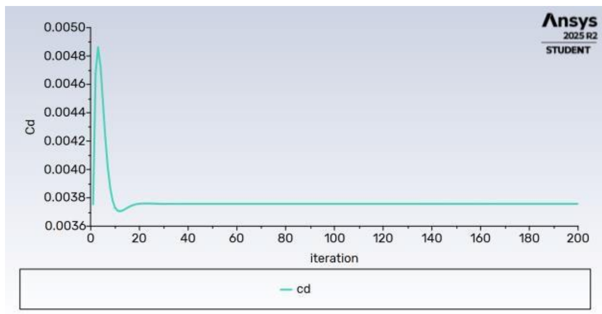

Figure 7b. Drag coefficient, -2° AOA.

3.3 0° Angle of Attack

For reference, the 0° case is shown again below alongside the other angles for direct comparison.

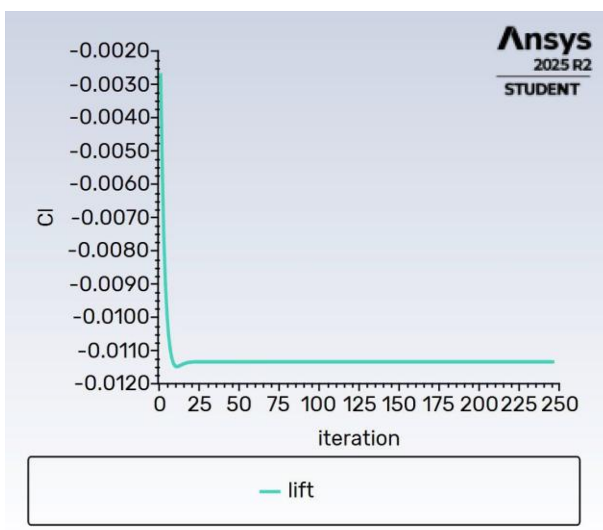

Figure 8a. Lift coefficient, 0° AOA.

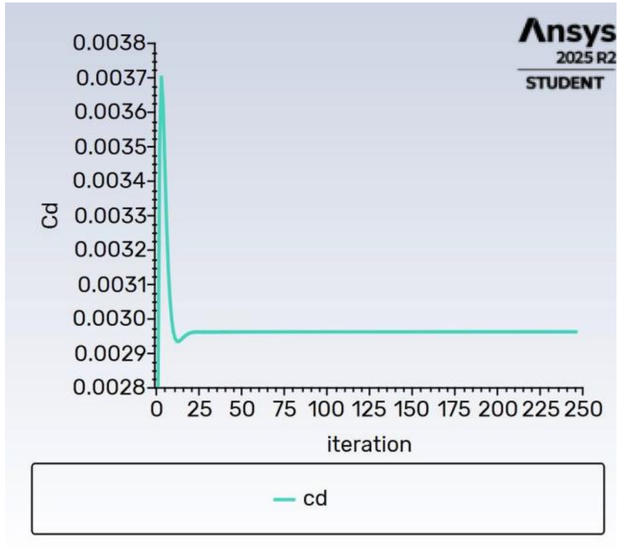

Figure 8b. Drag coefficient, 0° AOA.

3.4 2° Angle of Attack

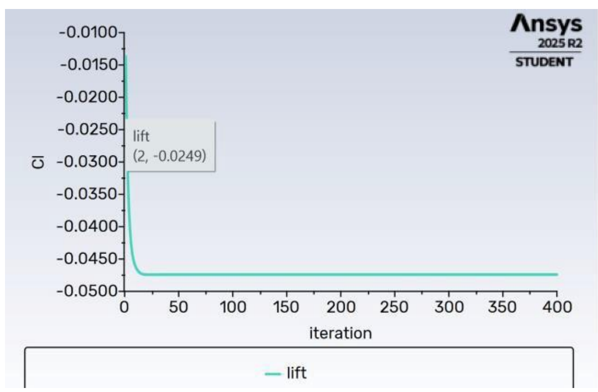

Figure 9a. Lift coefficient, 2° AOA.

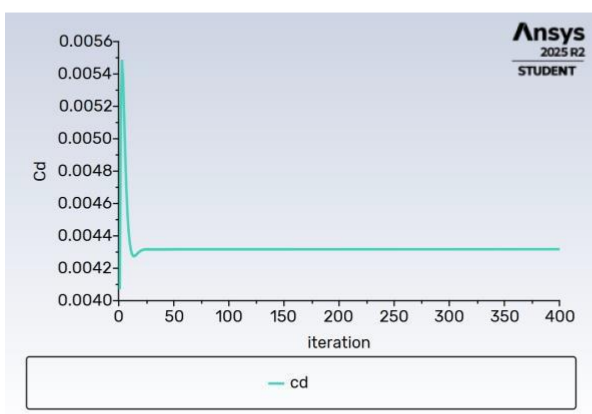

Figure 9b. Drag coefficient, 2° AOA.

3.5 4° Angle of Attack

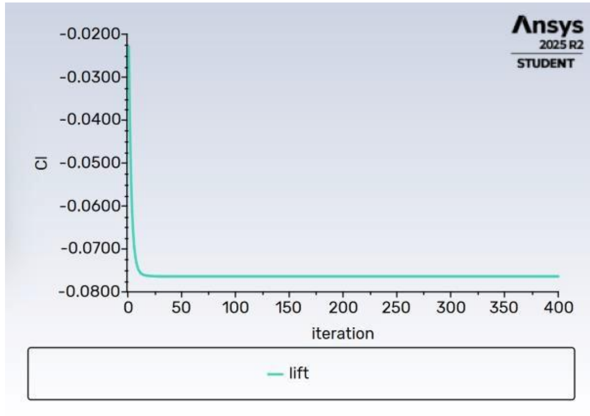

Figure 10a. Lift coefficient, 4° AOA.

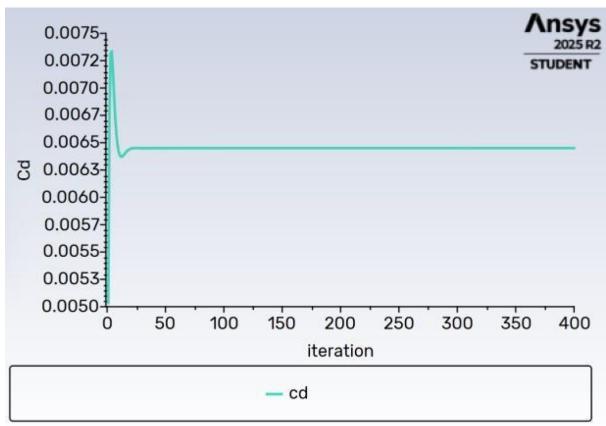

Figure 10b. Drag coefficient, 4° AOA.

Across the sweep, the drag coefficient increases steadily as the angle of attack increases, from roughly 0.00375 at -2° to 0.0065 at 4°. This trend is expected: as the airfoil is pitched further from its zero-lift orientation, the effective cross-section it presents to the oncoming flow increases, which increases pressure drag.

The lift coefficient follows the expected trend as well: Cl rises from roughly -0.0165 at -2° to +0.0113 at 0° and continues climbing to +0.045 at 2° and +0.072 at 4°. This matches the classical lift curve for a cambered airfoil such as the NACA 2412, lift increases steadily with angle of attack over this range, and even shows a small positive Cl at 0° angle of attack, consistent with the built-in camber of the airfoil generating lift even when the chord line is aligned with the freestream. Because drag also rises across the same range, the sweep captures the expected trade-off between lift and drag as angle of attack increases toward stall.

Conclusion

The CFD simulation of the Cessna 172's NACA 2412 main wing and NACA 0012 stabilizer successfully captured the expected qualitative flow features at 0° angle of attack: a region of low pressure and accelerated flow above the main wing that is consistent with lift generation. Extending the analysis across angles of attack from -2° to 4° showed the expected aerodynamic trends: lift coefficient increasing with angle of attack, alongside a steady increase in drag coefficient. This workflow (geometry, meshing, steady-state solution, and coefficient monitoring) provides a solid basis for evaluating the aerodynamic performance of the Cessna 172 airfoils across a range of flight conditions.

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